The key difference between the two layouts is that in the one figure 15.3, edges cross each other multiple times, making it difficult to follow them. In figure 15.2, instead, no edge was crossing: please meet a *planar embedding* for a *planar graph*! We'll define those in a minute, don't worry; before that, one further consideration: the drawing could get even worse, if you think about it – at least in figure 15.3 edges don't overlap with vertices here.

15.1.1 Some Basic Definitions

In the previous section, we have witnessed how drawing a graph without intersections between its edges makes the visualization a lot clearer. But, is it always possible to do so, to avoid these intersections?

Hold on to this question, we'll get back to it; meanwhile, we can give a couple of definitions that we'll use during this and later chapters.

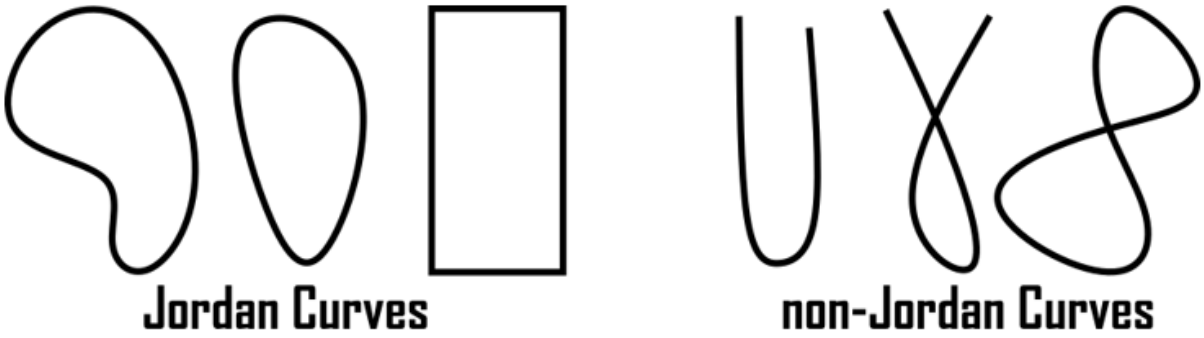


Figure 15.4 Examples of Jordan and non-Jordan curves. A Jordan curve is a closed curve whose border doesn't have intersection points (it's not "twisted" like the last two examples). Notice how a rectangle